INCH-POUND
MIL-PRF-52347D
25 March 1996
SUPERSEDING
MIL-S-52347C
21 November 1980

PERFORMANCE SPECIFICATION

SUSPENSION KIT, HOSELINE

This specification is approved for us by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a suspension kit for use in suspending a hoseline system over rough terrain (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other section of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank- automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 3835

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:15Z
Check the source to verify that this is the current version before use.

2.2 Government documents.

2.2.1 Specification, standards, and handbooks. The following specification, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

GGG-B-490	- Cleaning Methods for Ferrous, Surface and Pretreatments for Organic Coatings.
QQ-S-698	- Sheet Steel and Strip, Low Carbon.

STANDARDS

FEDERAL

FED-STD-595	- Colors Used in Government Procurement.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

DEPARTMENT OF DEFENSE

D13200E4792	- Saddle, Hose, Fabric.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:15Z
Check the source to verify that this is the current version before use.

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR QUALITY CONTROL (ASQC)

ANSI/ASQC Z1.4 - Sampling Procedures and Tables for Inspections by Attributes (DoD Adopted).

(Application for copies should be addressed to American National Standard Institute, 11 West 42nd Street, New York, NY 10036.)

AMERICAN STANDARDS AND TESTING MATERIALS (ASTM)

ASTM F1145 - Standard Specification for Turnbuckets, Swaged Welded Forged.

(Copies of ASTM publications may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshochocken, PA 19428-2959.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE HS1086 - Metals and Alloys in the United Numbering System (UNS).

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

2.4 Order of preference. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:15Z
Check the source to verify that this is the current version before use.

3.2 Materials. Materials used shall be in accordance with the manufacturer's materials specifications for the applicable items. The materials shall be capable of marking all of the operational and environmental requirements specified herein (see 4.5.1). Recovered materials shall be used to the maximum extent practicable.

3.2.1 Material deterioration prevention and control. The suspension kit shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the suspension kit may be exposed (see 4.5.2).

3.2.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion (see 4.5.2).

3.3 Design and construction. The suspension kit shall consist of a chest and the components listed in table I (see 4.5.2).

TABLE I. Chest components.

Component	Quantity
Wire rope	350 feet
Manila rope	350 feet
Wire-rope clamps	25
Shackles	60
Hose saddles	60
Steel pickets	14
Steel blocks	4
Turnbuckles	4
Wire rope thimbles	4

3.3.1 Chest. A suitable chest shall be provided for the components of the kit. The chest shall be not more than 22 inches in height. The chest shall have a hinged cover and latches to secure the closed position. The cover shall be self-supporting in the open position. Handles shall be provided. The handles shall of a size to enable lifting of the fully loaded chest by gloved personnel. The chest cover and front shall be labeled "HOSELINE SUSPENSION KIT" in 1 inch high letters. Bins, cradles, or blocking shall be provided for the components of the kit to prevent damage during shipment and handling. The chest shall be of sufficient strength to transport the kit components under all transport conditions.

3.3.1.1 Loading plan. A loading plan indicating the location of all items in the chest shall be furnished. The loading plan shall be permanently attached to the inside of the cover.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:15Z
Check the source to verify that this is the current version before use.

3.3.2 Kit components construction.

3.3.2.1 Wire rope. The wire rope shall be a 0.375 inch diameter, general purpose type. It shall have a nominal strength of 4080 pounds (lbs) and an acceptance breaking strength of 3980 lbs (see 4.5.3).

3.3.2.2 Manila rope. The manila rope shall be mildew-resistant treated 0.50 inch (in.) diameter, standard 3 strand type. It shall have a minimum breaking strength of 2650 lbs. As an alternate, an equivalent non-metallic material may be used (see 4.5.4).

3.3.2.3 Wire rope clamps. Wire rope clamps shall be the single grip, single saddle type, 0.375 in. size, with U-bolts and nuts. Configuration details of clamp components may be the contractor's design; however, the assembled clamp shall be representative of the illustration in figure 1. The inside width between the parallel U-bolt legs shall be sufficient to permit properly griping a rope double over. The clamps shall withstand, without distortion, 18.5 ft-lbs torque (see 4.5.5).

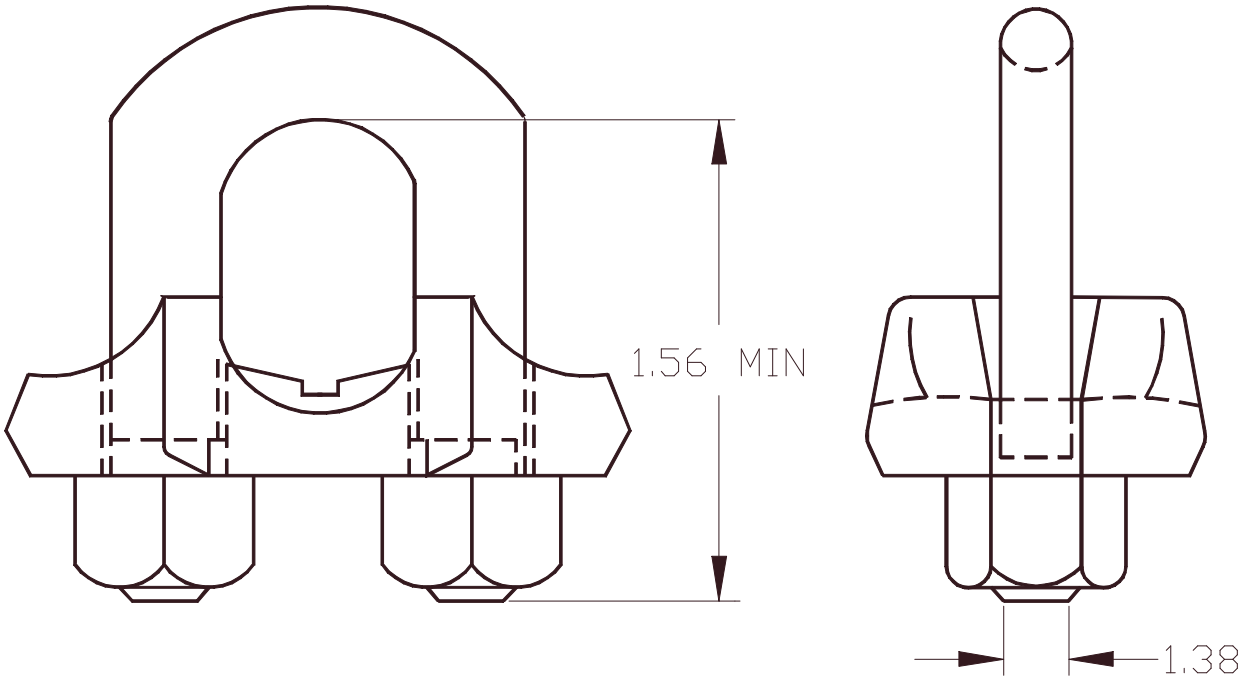


FIGURE 1. Wire rope clamp.

3.3.2.4 Shackles. Shackles shall be the screw pin anchor type, 0.375 in. size, similar to figure 2. The shackles shall have a safe working strength of 1590 lbs, a proof strength of 3180 lbs, and a minimum breaking strength of 7950 lbs (see 4.5.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:15Z
Check the source to verify that this is the current version before use.

3.3.2.5 Hose saddles. Hose saddles shall be constructed as shown on drawing D13200E4792.

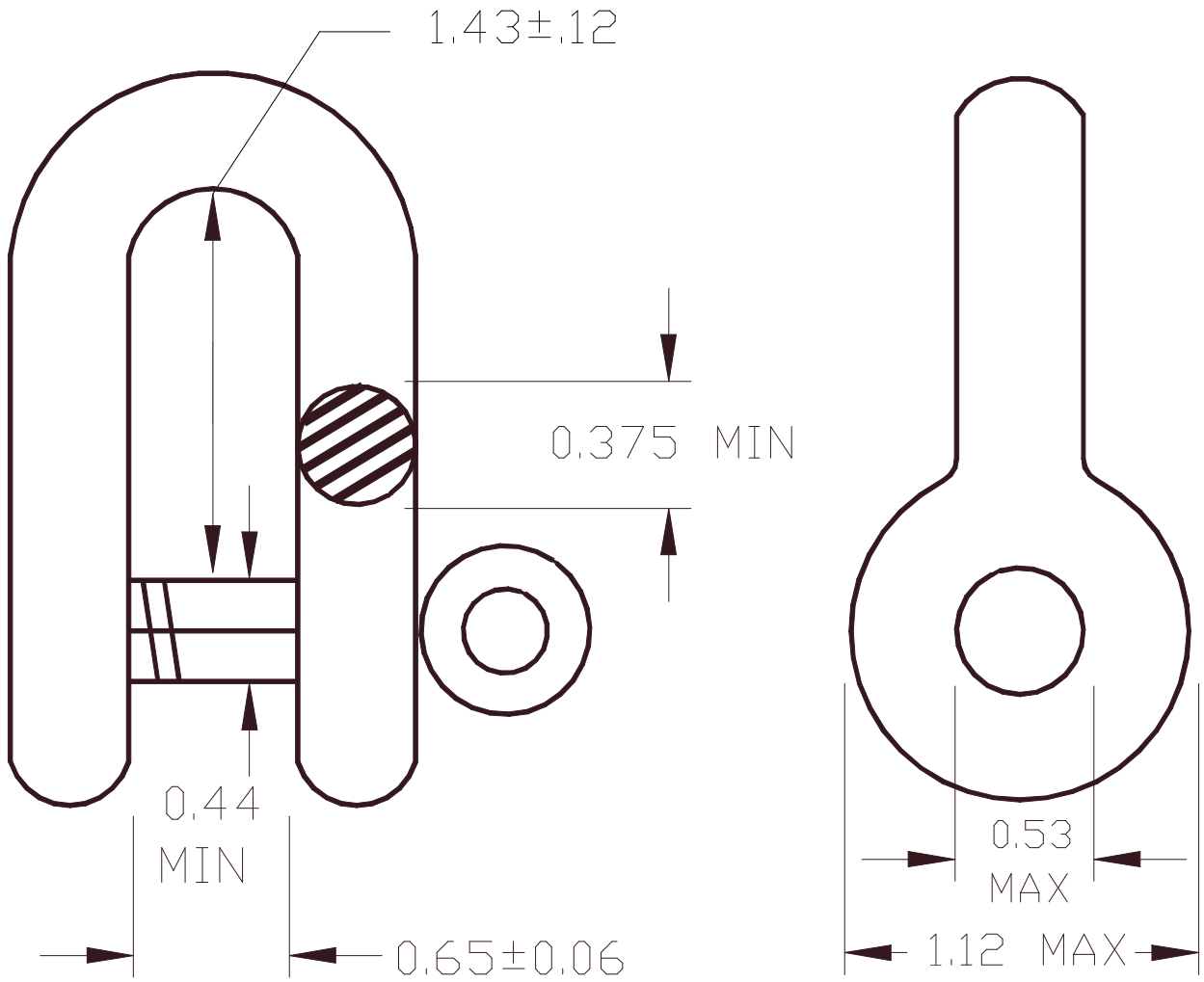


FIGURE 2. Shackle.

3.3.2.6 Steel pickets. The pickets shall be solid steel bars conforming to UNS G10200, UNS G10180 (see SAE HS1086), or equivalent strength material. The diameter shall be not less than 0.750 in. and the length shall be 35.75 in. ± 0.25 in. One end of each bar shall be sharpened to a 30-degree included angle point.

3.3.2.7 Steel blocks. The steel block shall conform to GGG-B-490, type II, single sheave, 4-inch shell length or commercial equivalent. Center pin diameter shall be not less than 0.360 in. Rig assemblies required for blocks shall be as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:15Z
Check the source to verify that this is the current version before use.

3.3.2.8 Turnbuckles. Turnbuckles shall conform to ASTM F1145, type I, grade 1, class E.

3.3.2.9 Wire rope thimbles. The wire rope thimbles shall be constructed of cold-rolled or hot-rolled steel conforming to QQ-S-698 or equivalent strength material. The wire rope thimbles shall be 0.375 in. size of split oval construction, semi-circular at the closed ends, with the sides tapering together at the free ends. The free ends shall have a rounded shape. Thimbles shall conform to the design shown in figure 3.

3.4 Treatment and painting. The portions of the kit normally painted shall be cleaned, treated, and painted in accordance with the materials being used. Unless otherwise specified (see 6.2), paint color shall be green 383, number 34094 of FED-STD-195.

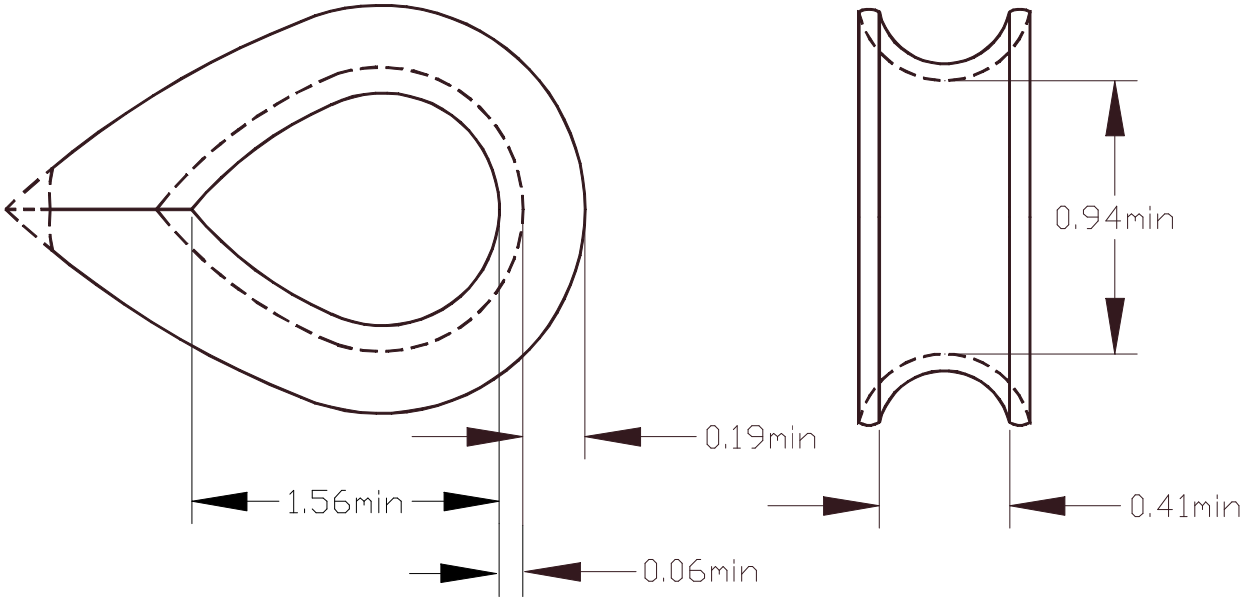


FIGURE 3. Wire rope thimble.

3.5 Identification marking. Identification marking shall be permanent and legible and shall include as a minimum, the manufacturer's name and part number and the National Stock Number (NSN). Special markings, if any, shall be as specified (see 6.2).

3.6 Workmanship. All components of the kit and chest shall be free of extraneous material such as sand, dirt, sprues, scale, and flux. External surfaces shall be free from burrs, sharp edges, and corners.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:15Z
Check the source to verify that this is the current version before use.

3.6.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not damage the material. Corners shall be square. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to ensure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly (see 4.5.2).

4. VERIFICATION

4.1 Classification of inspection. The inspection conditions specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. Unless otherwise specified (see 6.2), first article inspection shall be performed on preproduction or initial production samples as specified when a first article sample is required (see 3.1). This inspection shall include the examinations of 4.4 and the tests of 4.5.3 through 4.5.6.2 (see table II).

TABLE II. Classification of defects.

Category	Defect	Method of examination
Major:		
101	Material not as specified (see 3.2).	Visual
102	Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments (see 3.2.1).	Visual
103	Dissimilar metals not effectively insulated from each other (see 3.2.2).	Visual
104	Chest not as specified (see 3.3.1).	Visual
105	Loading plan missing (see 3.3.1.1).	Visual
106	Wire rope not as specified (3.3.2.1).	Visual
107	Manila rope not as specified (see 3.3.2.2).	Visual
108	Wire rope clamps not as specified (see 3.3.2.3).	SIE 1/
109	Shackles not as specified (see 3.3.2.4).	SIE
110	Hose saddles not as shown on drawing (see 3.3.2.5).	SIE
111	Steel pickets not as specified (see 3.3.2.6).	Visual
112	Steel blocks not as specified (see 3.3.2.7).	SIE
113	Turnbuckles not as specified (see 3.3.2.8).	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:15Z
Check the source to verify that this is the current version before use.

TABLE II. Classification of defects - Continued.

Category	Defect	Method of examination
114	Wire rope thimbles not as specified (see 3.3.2.9).	SIE
115	Workmanship not as specified (see 3.6).	Visual
116	Metal fabrication not as specified (see 3.6.1).	Visual
Minor:		
201	Treatment and painting not as specified (see 3.4).	Visual
203	Identification marking missing, incomplete or illegible (see 3.5).	Visual

1/ SIE = Standard Inspection Equipment.

TABLE III. Classification of inspections.

Title	Requirement	Inspec-tion	First article	Conformance tests
Wire rope tensile strength	3.3.2.1	4.5.3	X	
Manila rope breaking strength	3.3.2.2.	4.5.4	X	
Wire rope clamp strength	3.3.2.3	4.5.5	X	
Shackle proof strength	3.3.2.4	4.5.6.1	X	X
Shackle breaking strength	3.3.2.4	4.5.6.2	X	X

4.3 Conformance inspection. Conformance inspection shall consist of the examinations of 4.4 and the tests of 4.5.6.1 and 4.5.6.2.

4.4 Examination.

4.4.1 Sampling. Samples from an inspection lot for conformance inspection shall be selected in accordance with ANSI/ASQC Z1.4. Any redesign or modification of the contractor's standard to comply with specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements of requirements of 3.2 through 3.6 as listed in table II. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Method of inspection.

4.5.1 Materials. Conformance to 3.2 shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records,

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:15Z
Check the source to verify that this is the current version before use.

processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.5.2 Defects. Conformance to 3.2 through 3.6 shall be determined by examination for the defects listed in table III. Examination shall be visual, tactile, or by measurement with SIE.

4.5.3 Wire rope tensile strength test. To determine conformance to 3.3.2.1, the wire rope shall be subjected to the breaking strength load specified in the requirement. The wire rope shall not breaks at any load less than specified.

4.5.4 Manila rope breaking strength test. To determine conformance to 3.3.2.2, the manila rope shall be subjected to the breaking strength load specified in the requirement. The manila rope shall not break at any load less than specified.

4.5.5 Wire rope clamp strength test. To determine conformance to 3.3.2.3, two steel rods 0.375 in. in diameter shall be inserted one on top the other in the clamp opening. The clamp nuts shall be tightened to 18.5 ft-lbs torque. Tightening shall be accomplished progressively by applying torque alternately to each nut in increments of 10 percent to the maximum torque value. The maximum torque on each nut shall be held for 10 minutes (min). Following the 10 min period, the clamp shall be disassembled and visually examined for distortion, then tested by being reassembled and disassembled by hand without tools. Nonconformance to this test shall make the clamp defective.

4.5.6 Shackle strength tests.

4.5.6.1 Proof test. To determine conformance to 3.3.2.4, the shackle shall be subjected to the proof strength specified in the requirement. The diameter of the attachment used in the proof test shall be not greater than the pin diameter of the shackle being tested and the proof load shall be at the center of the pin. After releasing the proof load, the shackle shall be examined for surface rupture or defects that would interfere with serviceability or prevent disassembly of the pin.

4.5.6.2 Breaking and ductility tests. After proof testing, the shackle shall be subjected to the breaking strength specified in the requirement. Attachment and load application shall be shall be the same as in 4.5.6.1. Following the test, the shackle shall be examined for signs of failure.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:15Z
Check the source to verify that this is the current version before use.

packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The suspension kit is intended for us with 4 inches, light weight collapsible hoseline system.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and data of the specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
c. If first article is not required (see 3.1).
d. Rig assembly required for blocks (see 3.3.2.7).
e. Color required, if other then specified (see 3.4).
f. Any special marking (see 3.5).
g. Level of packing required (see 5.1).

6.3 Subject term (key word) listing.

Hose saddles
Pickets
Shackles
Wire ropes

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - AT

Preparing Activity:
Army - AT

Review Activity:
DLA - CS

(Project 3835-A111)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:15Z
Check the source to verify that this is the current version before use.